Electrochemistry

30 questions: 15 multiple choice, 15 structured.

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1 Core Electrolysis [1]

Which statement is the correct definition of electrolysis?

A The decomposition of an ionic compound, when molten or in aqueous solution, by the passage of an electric current

B The breakdown of a compound using heat energy alone

C The reaction of a metal with an acid to produce a salt and hydrogen

D The process by which a metal ore is converted into the pure metal by heating with carbon

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2 Core Electrolysis [1]

Molten zinc chloride is electrolysed using two graphite rods connected to a battery. What is the name given to the positive electrode?

A cathode

B anode

C electrolyte

D terminal

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3 Core Electrolysis [1]

Molten lead(II) bromide is electrolysed using inert electrodes. Which pair correctly describes the two products and the electrode at which each forms?

A lead at the anode, bromine at the cathode

B hydrogen at the cathode, oxygen at the anode

C lead at the cathode, bromine at the anode

D lead at the cathode, oxygen at the anode

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4 Core Electrolysis [1]

In the electrolysis of any molten ionic compound, which statement is always true?

A A non-metal element is always produced at the cathode

B Two different metals are produced, one at each electrode

C A metal, or hydrogen, is produced at the cathode, and a non-metal other than hydrogen is produced at the anode

D A metal is produced at the anode and a non-metal at the cathode

5 Core Electrolysis [1]

Dilute sulfuric acid is electrolysed using platinum electrodes. What are the products at the cathode and at the anode respectively?

A hydrogen; oxygen

B oxygen; hydrogen

C sulfur; oxygen

D hydrogen; sulfur dioxide

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6 Core Electrolysis [6]

Molten lead(II) bromide is electrolysed in a school laboratory using two graphite electrodes connected to a power supply.

(a) State which electrode – the anode or the cathode – is the negative electrode, and name the product that forms there. [2]

(b) State the product that forms at the anode. [1]

(c) Describe two observations that would confirm the identity of the product at the anode. [2]

(d) State the name of the molten substance that is decomposed in this experiment. [1]

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7 Core Electrolysis [4]

Molten lithium chloride is held in a crucible and electrolysed using two carbon electrodes connected to a d.c. power supply.

(a) Define the term *electrolysis*. [2]

(b) State the name given to:

(i) the electrode connected to the negative terminal of the power supply [1]

(ii) the molten lithium chloride [1]

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8 Core Electrolysis [1]

Stainless steel cutlery is often electroplated with a thin layer of chromium. What are the main reasons for doing this?

A to improve its appearance and increase its resistance to corrosion

B to make the cutlery lighter and cheaper to produce

C to increase its melting point so it can be sterilised at higher temperatures

D to make the cutlery magnetic

9 Core Electrolysis [5]

A steel spoon is to be electroplated with silver.

(a) State two reasons why the spoon is electroplated with silver. [2]

(b) Describe how the spoon could be electroplated with silver. Your answer should name which electrode the spoon should be, what the other electrode should be made of, and a suitable electrolyte. [3]

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10 Core Electrolysis [1]

Molten calcium bromide is electrolysed using inert electrodes. What are the products formed at the cathode and at the anode respectively?

A bromine; calcium

B calcium; oxygen

C calcium; bromine

D hydrogen; bromine

11 **Supplement** Electrolysis [1]

Aqueous copper(II) sulfate is electrolysed using two inert carbon electrodes. What is observed at the cathode?

A a colourless gas is released

B a pink/brown solid metal is deposited

C a green solid metal is deposited

D an orange-brown gas is released

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12 **Supplement** Electrolysis [7]

Aqueous copper(II) sulfate is electrolysed twice: once using two inert carbon electrodes, and once, separately, using two copper electrodes.

(a) State the product formed at the cathode. This is the same in both experiments. [1]

(b) With carbon electrodes, state the product formed at the anode. [1]

(c) With copper electrodes, describe what happens at the anode instead. [2]

(d) Explain why, when copper electrodes are used, the blue colour of the copper(II) sulfate solution stays roughly constant, but with carbon electrodes it gradually fades. [3]

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13 **Supplement** Electrolysis [1]

Molten sodium chloride is electrolysed using inert electrodes connected to a battery. In the external circuit (the wires connecting the battery to the electrodes), in which direction do electrons flow?

A from the cathode to the anode, through the wires

B electrons do not flow in the external circuit; only ions move

C from the cathode to the anode, through the electrolyte

D from the anode to the cathode, through the wires

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14 **Supplement** Electrolysis [6]

Molten magnesium chloride is electrolysed using inert graphite electrodes connected to a battery.

(a) Write the ionic half-equation for the reaction at the cathode. [2]

(b) Write the ionic half-equation for the reaction at the anode. [2]

(c) Describe how charge is carried:

(i) through the electrolyte [1]

(ii) through the external circuit [1]

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15 **Supplement** Electrolysis [1]

Concentrated aqueous potassium bromide is electrolysed using inert electrodes. What is formed at the anode?

A oxygen

B hydrogen

C bromine

D potassium

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16 **Supplement** Electrolysis [7]

Two separate electrolysis experiments are carried out using inert electrodes: Experiment 1 uses a dilute aqueous solution of sodium iodide; Experiment 2 uses a concentrated aqueous solution of sodium iodide.

(a) State the product formed at the cathode in both experiments, and explain why sodium metal is not formed there. [2]

(b) State the product formed at the anode in Experiment 1 (dilute solution). [1]

(c) State the product formed at the anode in Experiment 2 (concentrated solution). [1]

(d) Explain, in terms of the ions present, why the anode product is different between the two experiments. [3]

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17 **Supplement** Electrolysis [1]

Molten potassium chloride is electrolysed using inert electrodes. Which half-equation correctly represents the reaction at the anode?

A $K^+ + e^- \rightarrow K$

B $Cl_2 + 2e^- \rightarrow 2Cl^-$

C $2Cl^- + 2e^- \rightarrow Cl_2$

D $2Cl^- \rightarrow Cl_2 + 2e^-$

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18 **Supplement** Electrolysis [6]

(a) Molten lithium bromide is electrolysed using inert electrodes. Write balanced ionic half-equations for the reactions at:

(i) the cathode [1]

(ii) the anode [1]

(b) Dilute sulfuric acid is electrolysed using inert electrodes. Write balanced ionic half-equations, including state symbols, for the reactions at:

(i) the cathode [2]

(ii) the anode [2]

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19 Core Hydrogen–oxygen fuel cells [1]

What is the only chemical product formed when a hydrogen–oxygen fuel cell operates?

A water

B carbon dioxide

C water and carbon dioxide

D hydrogen peroxide

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20 Core Hydrogen–oxygen fuel cells [1]

Which two substances react together inside a hydrogen–oxygen fuel cell to generate electricity?

A hydrogen and nitrogen

B hydrogen and oxygen

C methane and oxygen

D hydrogen and carbon dioxide

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21 Core Electrolysis [1]

Concentrated aqueous sodium chloride is electrolysed using inert carbon electrodes. What gas is produced at the anode?

A oxygen

B chlorine

C hydrogen

D sodium vapour

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22 Core Electrolysis [1]

Molten strontium chloride is electrolysed using inert electrodes. What products form at the cathode and at the anode respectively?

A chlorine; strontium

B strontium; oxygen

C hydrogen; chlorine

D strontium; chlorine

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23 Core Electrolysis [5]

Dilute sulfuric acid is electrolysed in the laboratory using platinum electrodes.

(a) Name the gas produced at the cathode and describe a test that would confirm its identity, including the result. [2]

(b) Name the gas produced at the anode and describe a test that would confirm its identity, including the result. [2]

(c) State the electrolyte used in this experiment. [1]

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24 Core Electrolysis [5]

A car bumper made of steel is electroplated with a thin layer of chromium before sale.

(a) State two reasons why the bumper is electroplated with chromium rather than left as bare steel. [2]

(b) Describe how the bumper could be electroplated with chromium. Your answer should name which electrode the bumper should be, a suitable material for the other electrode, and a suitable electrolyte. [3]

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25 **Supplement** Electrolysis [8]

Impure copper is purified using electrolysis. A block of impure copper is used as the anode and a thin strip of pure copper as the cathode. Both are dipped into aqueous copper(II) sulfate.

(a) Write the ionic half-equation for the reaction at the cathode. [2]

(b) Write the ionic half-equation for the reaction at the anode. [2]

(c) Explain what happens to the mass of the anode and the mass of the cathode as electrolysis proceeds. [2]

(d) Explain why the concentration of the copper(II) sulfate solution stays approximately constant during this process. [2]

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26 **Supplement** Electrolysis [8]

In industry, concentrated aqueous sodium chloride (brine) is electrolysed continuously using inert electrodes. This is called the chlor-alkali process.

(a) State the products formed at the cathode and at the anode. [2]

(b) Write the ionic half-equation for the reaction at the anode. [2]

(c) Name the solution that remains after electrolysis, and state one industrial use of it. [2]

(d) Explain, in terms of the ions present, why chlorine rather than oxygen is formed at the anode, even though oxygen is normally the preferred product at a carbon anode in dilute aqueous solution. [2]

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27 **Supplement** Electrolysis [5]

Dilute sulfuric acid is electrolysed using inert platinum electrodes connected to a battery.

(a) Describe how charge is carried through the dilute sulfuric acid (the electrolyte). [2]

(b) Describe how charge is carried through the wires of the external circuit. [1]

(c) State what happens to hydrogen ions when they reach the cathode. [2]

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28 Core Hydrogen–oxygen fuel cells [4]

A hydrogen–oxygen fuel cell is used to power the electric motor of an experimental car.

(a) State the two gases that are supplied to the fuel cell. [1]

(b) State the only chemical product of the fuel cell reaction, and explain why the car's exhaust does not release carbon dioxide, unlike a petrol-engine car. [3]

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29 **Supplement** Hydrogen–oxygen fuel cells [6]

(a) State two advantages of using a hydrogen–oxygen fuel cell, rather than a petrol engine, to power a vehicle. [2]

(b) State two disadvantages of using a hydrogen–oxygen fuel cell, rather than a petrol engine, to power a vehicle. [2]

(c) Suggest one reason why hydrogen fuel cell vehicles are not yet as widely used as petrol-engine vehicles. [2]

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30 **Supplement** Hydrogen–oxygen fuel cells [6]

A car manufacturer advertises its hydrogen fuel cell car as "100% pollution-free, with zero emissions."

(a) Explain why this claim is accurate in terms of what comes out of the car's exhaust while it is being driven. [2]

(b) Evaluate whether the claim that the car is "100% pollution-free" is a fair description of its overall environmental impact, considering how the hydrogen fuel might be produced. [4]

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